# Week 2

## Introduction

In the rapidly evolving world of electric vehicles (EVs), the efficiency and reliability of charging infrastructure are paramount. At the core of every EV charger lies an AC-DC converter, a crucial component responsible for transforming the alternating current from the electrical grid into the direct current required to charge EV batteries. This week, we delved into the fundamental principles and advanced topologies of these converters, starting with basic diode bridge rectifiers and progressing to sophisticated Power Factor Correction (PFC) converters. Understanding these foundational circuits is essential for grasping the complexities of modern power electronics, as they lay the groundwork for designing robust and efficient EV charging systems that meet stringent power quality standards. We explored the evolution from simple, uncontrolled rectification to active power factor correction, highlighting the trade-offs and design considerations that drive innovation in this critical field.

## Core Concepts

| Term | Definition | Significance |
| --- | --- | --- |
| `Single-Phase Full-Bridge Rectifier` | An AC-DC converter using four diodes to convert both positive and negative half-cycles of an AC input into a pulsating DC output. | A more efficient and common rectification method, providing continuous power delivery to the load compared to half-bridge. |
| `Pulsating DC` | A direct current or voltage that, while always having the same polarity, varies significantly in magnitude over time, typically following a rectified sinusoidal pattern. | Characterizes the output of basic rectifiers, indicating the need for filtering to achieve a smooth, constant DC voltage. |
| `Capacitive Filter` | A capacitor connected in parallel with the load at the output of a rectifier circuit. | Stores energy during peak voltage periods and releases it during voltage dips, significantly reducing ripple |

| Term | Definition | Significance |
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| | | and smoothing the DC output voltage. |
| **Ripple Voltage (ΔVo)** | The peak-to-peak variation in the DC output voltage of a rectifier with a filter. | A key performance metric; a smaller ripple indicates a smoother, more stable DC output, essential for sensitive loads. |
| **Peaky Input Current** | The characteristic waveform of the input current drawn from the AC source by a rectifier with a capacitive filter, featuring sharp, narrow pulses. | Indicates high harmonic content and a poor power factor, critical for grid interaction and efficiency. |
| **Power Factor (PF)** | A measure of how effectively electrical power is being used, defined as the ratio of real power to apparent power. | A critical power quality metric; non-unity PF leads to increased losses and harmonic distortion in the grid. |
| **Power Factor Correction (PFC)** | A technique used to improve the power factor of an AC-DC converter, making the current drawn from the source more sinusoidal and in phase with the voltage. | Essential for meeting regulatory standards and improving the overall efficiency and quality of power drawn from the grid. |
| **Boost Converter** | A DC-DC converter that steps up a DC input voltage to a higher DC output voltage, typically using an inductor, a switch, a diode, and a capacitor. | Utilized in PFC applications to regulate the output DC voltage and shape the input current. |
| **Duty Ratio (D)** | The fraction of a switching period during which a power switch is turned on, a key | Allows for the control of the average voltage across the inductor and, consequently, |

| Term | Definition | Significance |
|---|---|---|
| | control parameter in switched-mode power converters. | the average current flowing through it, crucial for PFC. |

## Detailed Analysis

### 1. Revisiting Diode Bridge Rectifier Fundamentals

In **Revisiting Diode Bridge Rectifier - II**, we began by contrasting the single-phase half-bridge rectifier with the single-phase full-bridge rectifier. The half-bridge, using a single diode, delivers power only during one half-cycle, resulting in a highly pulsating and discontinuous output. This highlights its limitations for continuous power delivery.

The single-phase full-bridge rectifier, however, employs four diodes in a bridge configuration to deliver power to a resistive load during both positive and negative half-cycles of the AC input.
* **Positive Half-Cycle (V_s > 0):** Diodes D1 and D4 are forward-biased, allowing current to flow through the load. The output voltage (V_d) equals the instantaneous input voltage (V_s).
* **Negative Half-Cycle (V_s < 0):** Diodes D2 and D3 are forward-biased. Current flows through the load in the same direction as the positive half-cycle, making V_d equal to -V_s (effectively rectifying the negative half-cycle to positive).
* **Waveforms:** The input is sinusoidal AC. The output voltage (V_d) is a full-wave rectified sinusoidal waveform, always positive but pulsating. For a purely resistive load, the source current (i_s) is sinusoidal and in phase with the input voltage, indicating a unity power factor.
* **Performance Metrics:** The average output voltage (V_d,avg) is `2 * V_s,peak / π`, which is double that of a half-bridge.
* **Diode Selection:** Each diode must withstand a Peak Repetitive Reverse Voltage (VRRM) equal to `V_s,peak`. The Average Forward Current (IF,avg) is `V_s,peak / (π * R_L)`, and the RMS Current (I_rms) is `V_s,peak / (2 * R_L)`.
* **Key Inferences:** While providing continuous power and unity power factor for resistive loads, the output voltage remains uncontrolled and pulsating, directly proportional to the peak input AC voltage.

### 2. Enhancing Rectifiers with Capacitive Filters

The limitations of the basic full-bridge rectifier, particularly its pulsating DC output, were addressed in **Revisiting Diode Bridge Rectifier with Capacitive Filter - I** and **Revisiting Diode Bridge Rectifier with Capacitive Filter - II**. A capacitive

filter, connected in parallel with the load, is introduced to smooth out the output voltage ripple.

- **Circuit Operation:** The capacitor charges when the input voltage exceeds the capacitor voltage (diodes conduct) and discharges through the load when the input voltage drops below the capacitor voltage (diodes are reverse-biased). This charge-discharge cycle significantly reduces the output voltage ripple, making it vary between a maximum (V_d,max) and minimum (V_d,min) value, rather than fluctuating to zero.
- **Peaky Input Current:** A significant drawback is the "peaky" nature of the input current (i_s). Current is drawn from the AC source only in short, sharp pulses when the capacitor is charging near the peaks of the input voltage. This non-sinusoidal current waveform indicates high harmonic content and a poor power factor.
- **Capacitor Sizing:** The capacitance value (C) is crucial for ripple control. It's calculated by balancing the energy gained during charging and lost during discharging. The formula derived is:
  `C = (1 - t_c / (T/2)) * P_L / (2 * V_0 * ΔV_0 * f_s)`
  where `t_c` is the charging time, `P_L` is load power, `V_0` is nominal output voltage, `ΔV_0` is ripple voltage, and `f_s` is supply frequency. Minimizing ripple (smaller `ΔV_0`) requires a larger capacitance and leads to a shorter charging time (`t_c`).
- **Component Ratings:**
- **Capacitor:** Beyond capacitance, its voltage rating (at least `V_d,max`) and RMS current handling capability (I_c,rms) are critical due to the peaky currents.
- **Diodes:** The Peak Repetitive Reverse Voltage (V_RRM) for each diode is `V_s,peak`. The Peak Forward Current (I_F,peak) for each diode is equal to the peak source current (I_s,peak), which can be very high. The Average Forward Current (I_F,avg) is `I_s,peak * t_c / T`, and the RMS Forward Current (I_F,rms) is `I_s,peak * sqrt(t_c / T)`.
- **Disadvantages:** The diode bridge rectifier with a capacitive filter inherently suffers from a non-unity power factor due to distorted input current, requiring oversizing of components to handle high peak currents. This motivates the need for more advanced solutions.

### Introduction

The limitations of passive rectification led to the development of Power Factor Correction (PFC) converters, introduced in **Single-phase Boost PFC Converter - I** and **Single-phase Boost PFC Converter - II**. These converters actively shape the

input current to be sinusoidal and in phase with the input voltage, achieving unity power factor while regulating the output DC voltage.

- **Architecture:** The single-phase boost PFC converter uses a cascaded structure:
- **Full-Bridge Diode Rectifier:** Converts the AC input `V_s = V_s,pk sin(ωt)` into a pulsating DC voltage `v_d = |V_s,pk sin(ωt)|`. This `v_d` serves as the input to the boost converter.
- **Boost Converter Stage:** Consists of an inductor (L), switches (S1, S2), and an output capacitor (C) with the load (R_L). The key is to actively control switches S1 and S2 to manipulate the voltage `V_converter` and shape the inductor current (i_L).
- **High Switching Frequency Assumption:** A crucial assumption is that the switching frequency ( `f_sw` ) is much higher than the AC supply frequency ( `f_s` ). This allows `v_d` to be considered nearly constant over a single switching period, simplifying control.
- **Boost Converter Operation:**
- **S1 ON (Duty Ratio D * T_s):** Inductor voltage `v_L = v_d`. Inductor current `i_L` increases.
- **S2 ON ((1 - D) * T_s):** Inductor voltage `v_L = v_d - V_output`. Inductor current `i_L` decreases (since `V_output > v_d` ).
- **Duty Ratio Derivation for PFC:** The duty ratio `D(t)` must vary continuously over the AC half-cycle to achieve both unity power factor and constant output voltage. Using volt-second balance across the inductor, the duty ratio is derived as:
  `D(t) = 1 - (v_d(t) / V_o)`
  where `V_o` is the regulated output voltage. This implies that `V_o` must always be greater than `V_s,pk` (the peak input AC voltage) for the boost converter to operate correctly.
- **Inductor Sizing:** The inductor (L) is sized to manage the ripple current (Δi_L). The maximum ripple current (Δi_L,max) occurs when `v_d(t) = V_o / 2`. The required inductance is calculated as:
  `L = V_o / (4 * Δi_L,max * f_sw)`
  This formula allows designers to limit current ripple to an acceptable level, ensuring stable operation and preventing excessive losses.

## Key Takeaways

- [ ] Single-phase full-bridge rectifiers provide continuous power delivery and unity power factor for purely resistive loads, but the output voltage is uncontrolled and pulsating.
- [ ] The average output voltage of a full-bridge rectifier is `2 * V_s,peak / π`, and each diode must withstand `V_s,peak` reverse voltage.
- [ ] Adding a capacitive filter to a full-bridge rectifier reduces output voltage ripple, providing a smoother DC output.
- [ ] Capacitive filters cause the input current to be "peaky" and non-sinusoidal, leading to a poor power factor and requiring oversized components.
- [ ] Capacitor sizing depends on load power, desired ripple voltage, nominal output voltage, and supply frequency, with RMS current handling being a critical selection criterion.
- [ ] Diode ratings in a filtered rectifier are significantly impacted by the high peak source currents drawn during capacitor charging.
- [ ] Power Factor Correction (PFC) converters are necessary to achieve unity power factor and regulated output voltage simultaneously, overcoming the limitations of simple rectifiers.
- [ ] The single-phase boost PFC converter uses a full-bridge rectifier followed by an active boost stage to shape the input current and regulate the output.
- [ ] The duty ratio `D(t)` in a boost PFC converter must vary dynamically with the instantaneous input voltage, calculated as `D(t) = 1 - (v_d(t) / V_o)`.
- [ ] For a boost PFC converter, the output DC voltage `V_o` must always be greater than the peak input AC voltage `V_s,pk`.

## Conclusion

This week's exploration has taken us from the fundamental principles of AC-DC conversion with diode bridge rectifiers to the sophisticated realm of active Power Factor Correction. We've seen how basic rectifiers, while simple, present significant challenges in terms of output voltage ripple and, when filtered, introduce severe power quality issues like peaky input currents and poor power factor. These limitations are particularly critical in high-power applications such as EV charging, where grid compatibility and efficiency are paramount.

The introduction of the single-phase boost PFC converter represents a crucial step forward. By actively controlling the switching elements, this topology can simultaneously achieve a unity power factor, drawing a clean, sinusoidal current

from the grid, and provide a stable, regulated DC output voltage. Understanding the dynamic duty ratio control and the principles of inductor sizing are key to designing these efficient and grid-friendly converters. This foundational knowledge is indispensable for engineers developing the next generation of power electronics, ensuring that our charging infrastructure is not only powerful but also sustainable and compliant with evolving energy standards. The journey continues with further analysis of component selection and control strategies to optimize these advanced power conversion systems.